

IoT Based Water Level and Water Quality Monitoring System for Water Tanks

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OVERVIEW

This proposed system is of an IoT based system to Manage and maintain water tanks in Domestic houses and small scale industries.

This system will provide a centralized system to manage and monitor status or water levels and water quality from any location at real time.

With the use of this system users can not only manage their Time efficiently but also overcome a number of technical issues related to management of water tanks.

INTRODUCTION

This proposed system to manage water quality and water levels of tanks.

This system can monitor the status of water levels in water tanks in real time and check the quality of water and display them in a centralized web application.

This system has several features such as,

- Automatic alerting of users regarding water levels
- Controlling or displaying status of the pumps
- Checking status of the water in the tank.

This system will be built as two separate components, one component for measuring water levels and another to check for water quality.

PROBLEM IDENTIFICATION

People are unaware of the water levels of tanks and fill tanks until they overflow causing major water wastages.

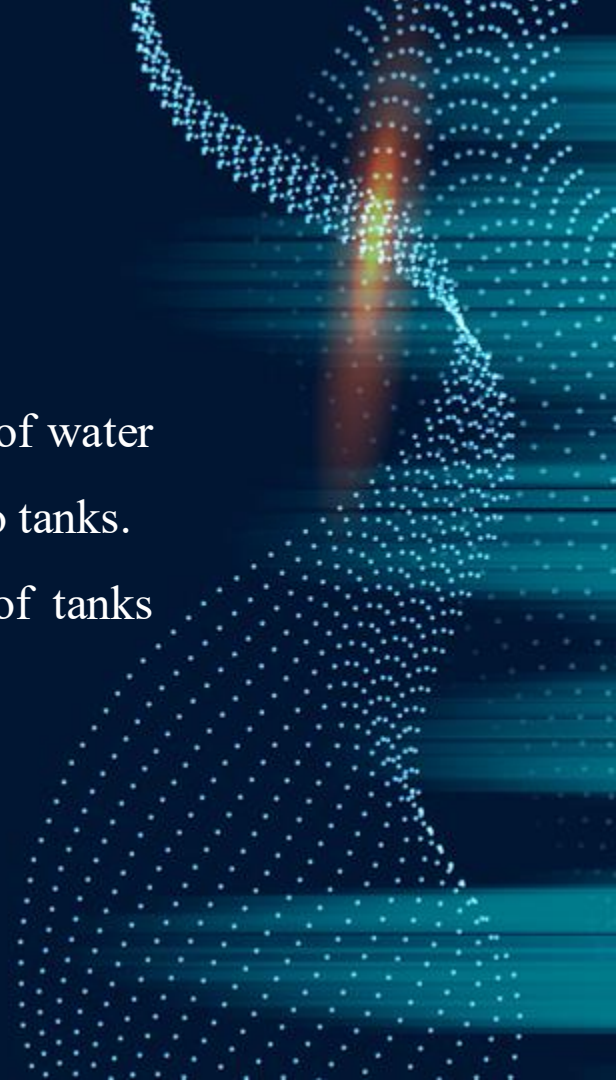
Dry running of pumps due to low water levels present in tanks, high water pressure differences due to excess pumping of water can cause major problems.

Due to the busy life styles of people they do not have time to wait for tanks to be filled.

Detecting water contamination in tanks which store water is a problem when different water sources are used.

OBJECTIVES

- Prevent wastage of water by at least 60%.
- Provide a centralized system for users to view status of water levels, quality of water and pumping status of water to tanks.
- Make pumping of water to tanks and view status of tanks easier.
- Ensure that uncontaminated water is stored in tanks



RESEARCH QUESTIONS



ACCESSIBILITY

Is the proposed System accessible for domestic households and small-Scale industries.



SYSTEM

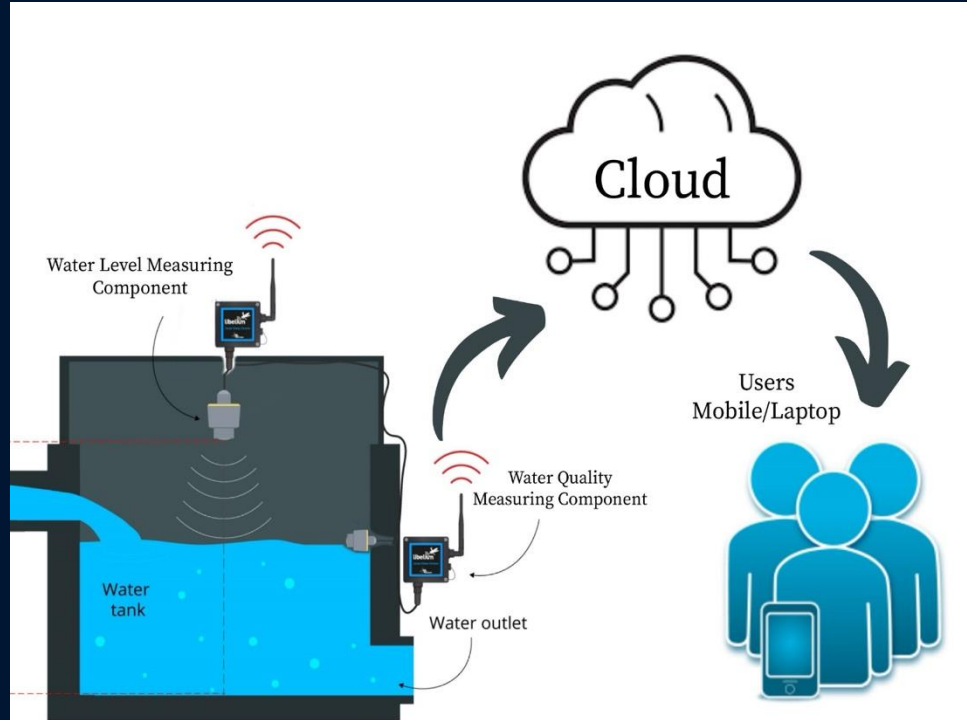
What aspects of water quality measurement have the most importance and what could be included for this proposed system?



TECHNOLOGIES

What fog computing systems and technologies could be used to automate a water quality and water level management system?

PROPOSED SOLUTION



Accessibility Diagram

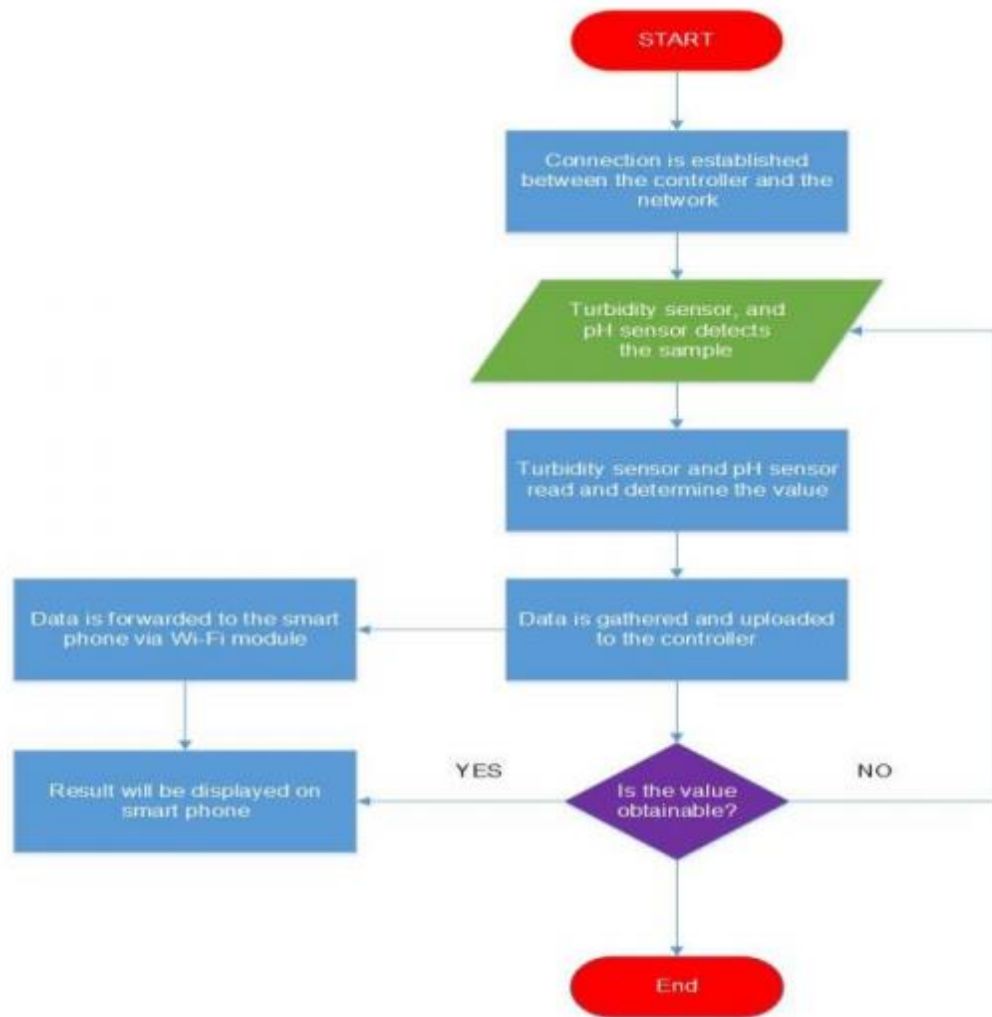
REQUIREMENTS

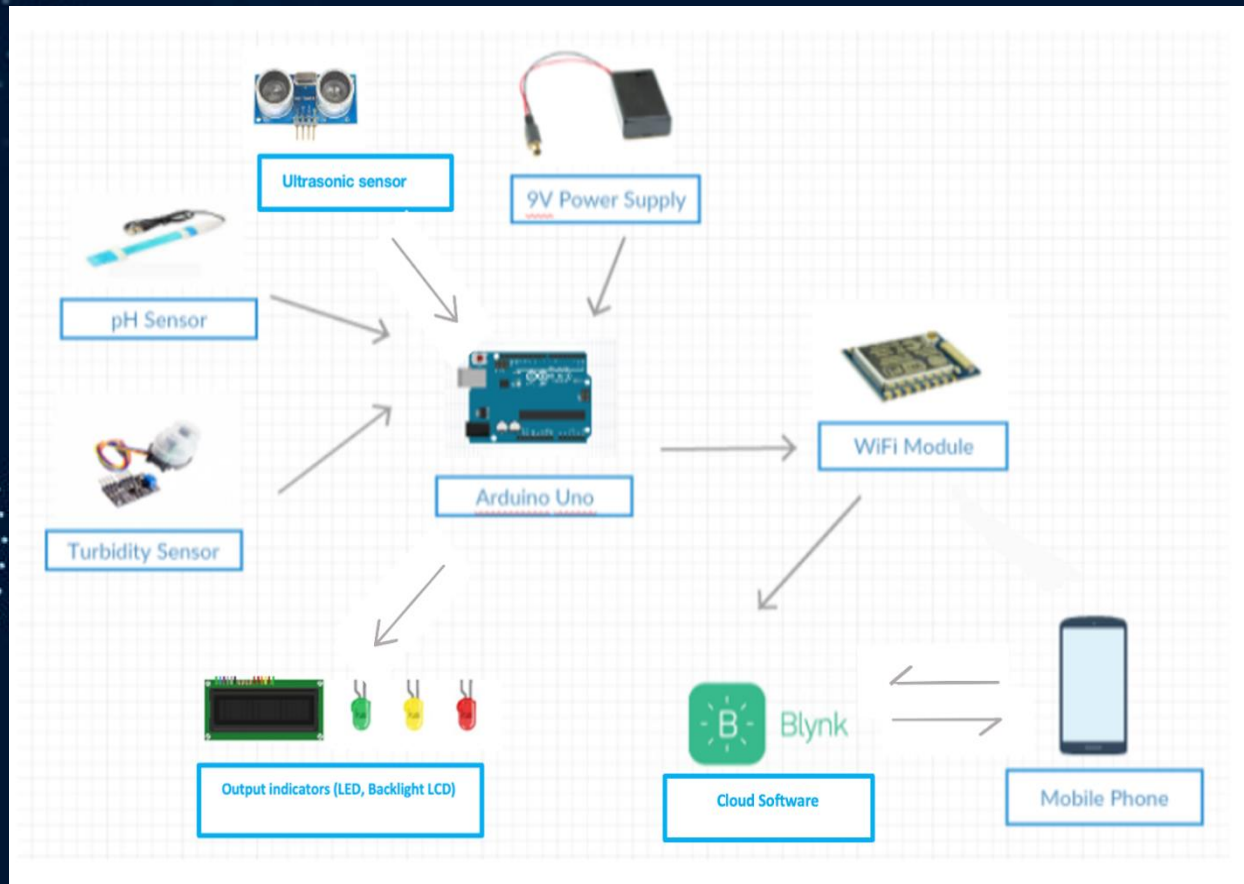
Listed below are the requirements for users to successfully set up the system,

- Each house or company must have a suitable water tank
- Users must register to the system before using it.
- Users must have a suitable device to access the web application.
- Users must have suitable supply of water or electric power.

Listed below are the main requirements of the system,

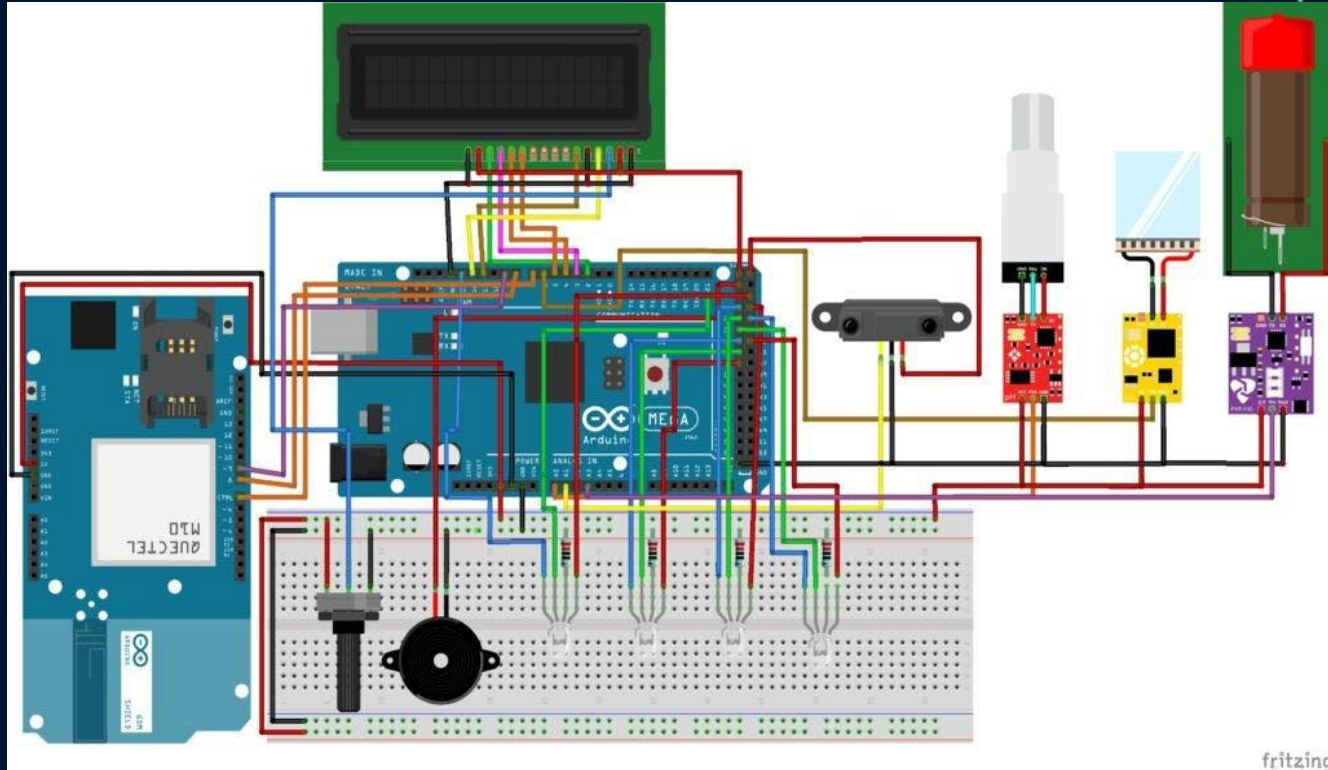
- The system must indicate proper water levels and water quality measurements at real time from any location.
- The system must display pump status whether water is being pumped or not and provide alerting messages to users if water level is exceeding maximum capacity or is below required levels.
- The pumps should be able to switch on or off using the system.





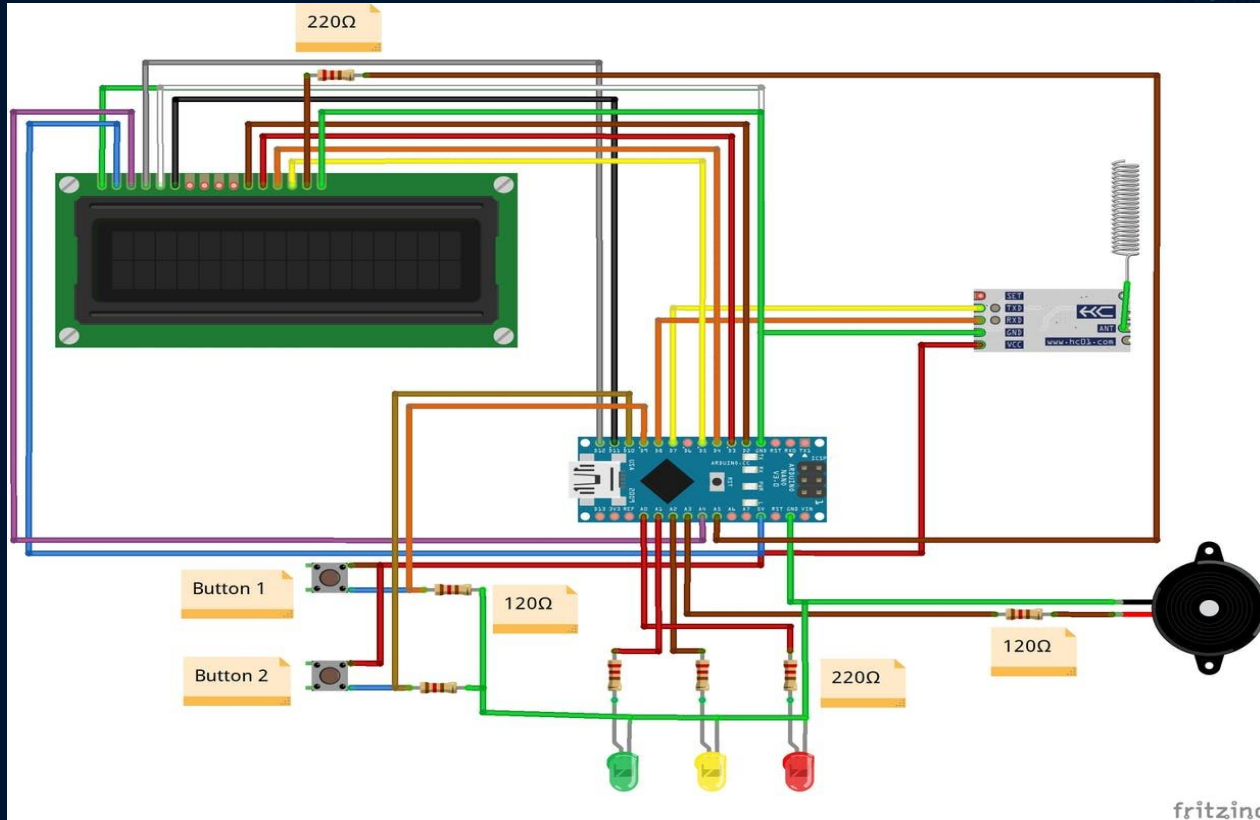
Water level and quality measurement system Accessibility diagram

CURCUIT DIAGRAMS



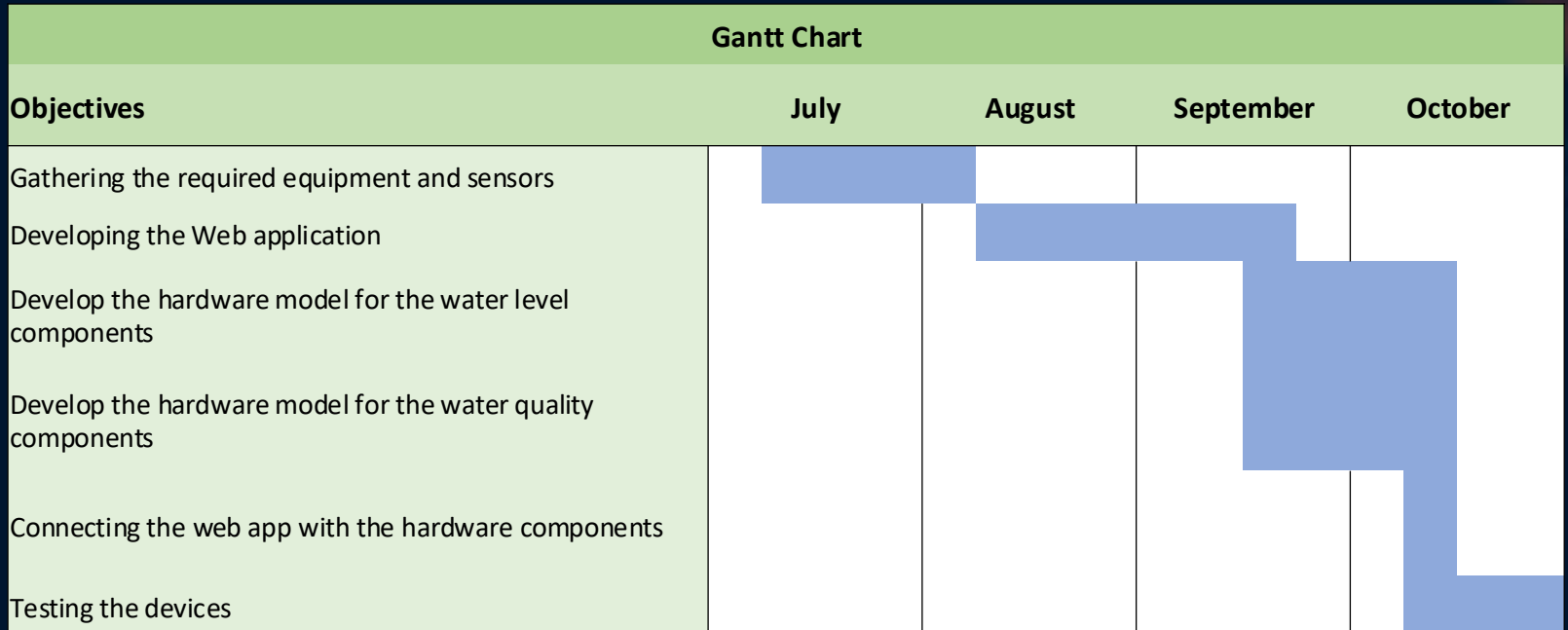
Circuit Diagram of
Water Quality
Measuring Component

CURCUIT DIAGRAMS



Circuit Diagram of Water Level Measuring Component

GANTT Chart



TESTING & EVALUATION

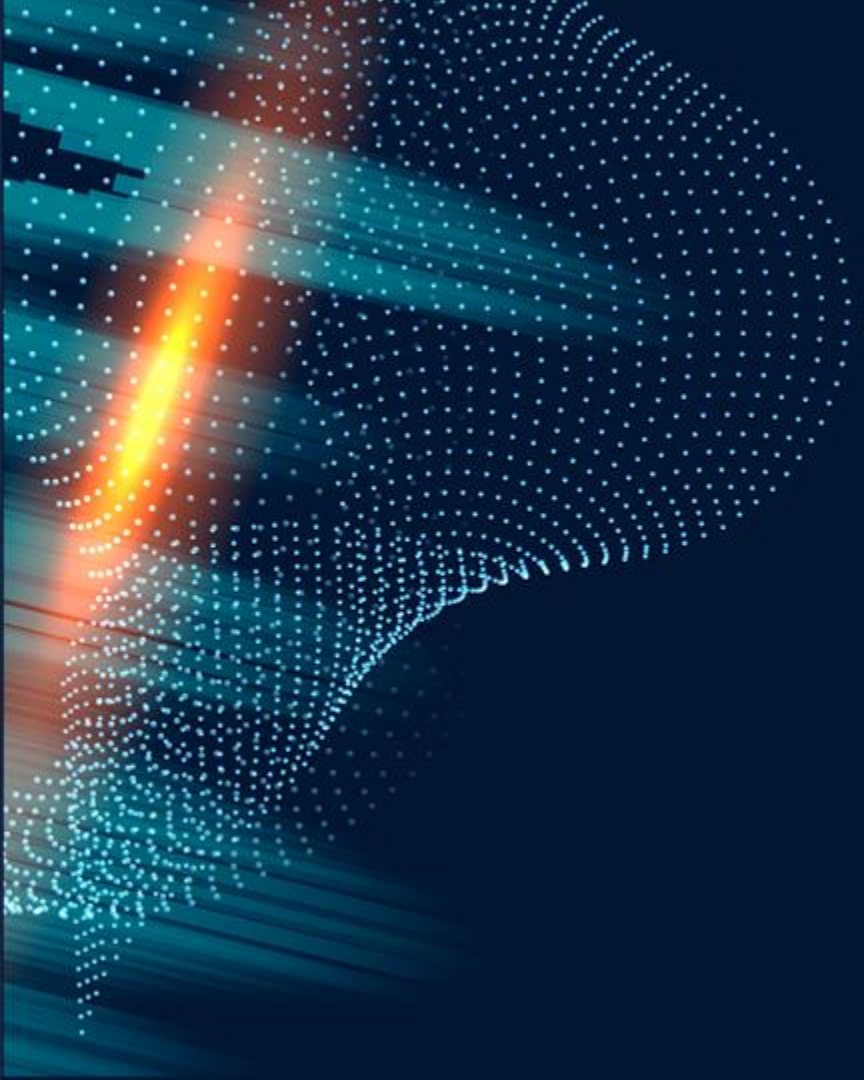
Testing

This project will be tested on 2 stages.

- Stage 1- Testing of Individual Components & web app
- Stage 2- Testing overall functionality of system

Evaluation

- The system will be given to 5 domestic house owners or small-scale industry owners who have water tanks for 3 days and will evaluate the product on his personal ideas and thoughts and overall functionality of the system. This includes his satisfaction towards overall use of the system, helpfulness and functionality, ease of use and whether the requirements of the user have been met.

An abstract background on the left side of the slide. It features a series of bright blue and orange light rays emanating from the left edge, creating a sense of depth and movement. Overlaid on these rays is a trail of small, glowing blue dots that curve and spiral outwards, resembling a particle path or a stylized wave. The overall color palette is dominated by deep blues and vibrant oranges.

Thank You!

Any Questions?